

Review:

Aerial Manipulation Using Multirotor UAV: A Review from the Aspect of Operating Space and Force

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Aerial manipulation: physical interaction with the environment by using a robotic manipulator attached to the airframe of an aerial robot. In the future one can expect that aerial manipulation will greatly extend the range of possible applications for mobile robotics, especially multirotor UAVs. This can range from inspection and maintenance of previously hard to reach pieces of infrastructure, to search and rescue applications. What kind of manipulator is attached to what position of the airframe is a key point in accomplishing the aerial robot's function and in the past, various aerial manipulation solutions have been proposed. This review paper gives an overview of the literature on aerial manipulation that have been proposed so far and classifies them by configuration of the workspace and function.

Keywords: multirotor UAV, aerial manipulation, tasks in high altitude, physical interaction

1. Introduction

Multirotor unmanned aerial vehicles (UAVs) have enjoyed rising popularity in the industrial and consumer industry in the recent 10 years [1, 2]. They are now commonly used not only for hobbyist applications, but also for civil engineering surveys, infrastructure inspections, monitoring, security, agriculture, logistics, and many other fields of the industry. Main applications of currently deployed multirotor UAVs are aerial photography or surveillance and transportation of various goods. In contrast to this, recently multiple research groups have been developing solutions aiming at manipulation of objects at high altitudes. This is largely due to the beneficial characteristics of multirotor UAVs, one of the currently predominantly used aerial robot platform. These multirotor UAVs are mechanically simple, have relatively simple control laws, can quickly move to high altitudes and are able to take-off and land vertically from the ground which is, in short, called vertical take-off and landing (VTOL). Since high-altitude work such as maintenance of bridges, tunnels, wind turbines, dam walls, and high-rise buildings can pose a danger to human workers, replacement and as-

sistance by aerial robots is a promising application. The technique of manipulating objects and physical interaction using a body in flight is called "aerial manipulation," and many studies have been conducted so far [3]. Our research group has developed many different types of multirotor aircrafts for aerial manipulation as well, giving us substantial experience and insight into this field.

In case of a UAV as an aerial robot, to perform an aerial manipulation task, it physically contacts the task object while flying. If a force is applied to the task object, the resulting reaction force will act on the aerial robot floating in the air significantly different than, for example, a mobile robot with solid contact to the ground. Also, positioning of the main robot frame relative to the attached manipulator's workspace is not trivial for an aerial robot to achieve within even an imprecise accuracy of several cm, due to the underlying fluid dynamics of the UAV's kinematics. Furthermore, since the maximum weight that can be mounted on the airframe, or in other words its maximum payload, is usually not very large, there are restrictions on the manipulation devices that can potentially be used, which can also be a major limitation factor. We believe that task specific, light weight manipulators with integrated sensing and onboard processing are one of the key technologies to realize various aerial manipulation tasks under such constraints. This paper introduces the configurations of various aerial robot research examples for aerial manipulation, and discusses the current issues and expectations for future development.

Previous review papers on aerial manipulation have focused on modeling and control [3, 4], and multiple vehicles [5]. In this paper, we review literature in aspect of their configuration of workspace, of force that is applied to the target, and discuss future challenges in aerial manipulation.

2. Workspace for Aerial Manipulation

In this work, we propose a classification of aerial manipulation based on workspace. The location, size, and shape of the workspace is depending on the desired task to be solved. In order to realize the desired workspace, it is first necessary to consider where to install the manipulator in the multirotor body. Due to the specific structure of

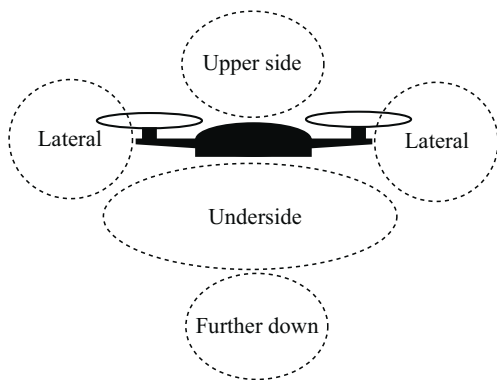


Fig. 1. Categorization of the workspaces in aerial manipulation.

a multirotor, the workspace above and below the airframe are divided by spinning propellers. This makes the lower workspace usually unreachable by an above mounted manipulator and vice versa. Furthermore a manipulator on the side of the airframe needs to either be extremely lightweight or have a special counterweight mechanism as to not imbalance the airframe.

In this work, the workspaces that can be reached with a manipulator equipped with the multirotor airframe are classified into underside, lateral, upper side, and further down of the airframe (**Fig. 1**).

2.1. Underside of the Airframe

A configuration most commonly seen is to attach a gripper or robot arm to the underside of the airframe to make the underside of the airframe a workspace. This is seen in many studies that have been published especially at the early stage of the aerial robotics research field [6–18]. For example, Mellinger et al. have developed small quadrotors equipped with several kinds of underside grippers and demonstrated the ability to grasp and manipulate a number of items such as as pieces of wood [6]. Pounds et al. have presented a solution for payload transportation using a single rotor helicopter with an underside located gripper [11]. Also, Huber et al. have reported a single rotor platform for aerial manipulation with a 7 degrees of freedom industrial robot arm, attached to the underside of the aerial platform's frame and achieved to grasp a pole standing on the ground [12]. In more recent research, aerial platforms evolve towards the capability of employing a wider workspace by equipping manipulators with a wide range of degrees of freedom, often used for various tasks such as infrastructure inspection and pick up of objects from narrow spaces [19–22].

The number of robot arms installed on aerial platforms has increased subsequently from one to two [23–27, a], and recently even platforms equipped with three manipulators have been developed. In [23], a human size dual arm system with compliant joints designed for aerial manipulation was presented. The aerial robot in [a] has dual arms and exhibits picking up the object and perching on the handrail. In [24], dual multi-degree of freedom manipulators were developed for a valve turning task using



Fig. 2. Aerial manipulator with three arms [30] (©2019 IEEE).

an aerial vehicle. [28, 29] proposed aerial robots with four legs, which are used as landing gear. Although not manipulators, it is an interesting concept to increase the functionality of an aerial robot by installing an even larger number of extremities.

Our research group has developed an aerial robot as well, that uses three robot arms attached to the lower part of the body for both “hands” to lift objects and “legs” to land (**Fig. 2**) [30, 31]. By using the same mechanism for various purposes, we show that even an aerial robot with a limited payload can be made multi-functional. By properly controlling the attitudes of the three robot arms, it is possible to land even on widely irregular terrain. The developed aircraft is equipped with a stereo camera at the bottom of the aircraft. By measuring the shape of the ground near the landing point, the ground contact positions of the tips of the three arms and the required attitude are determined using an onboard processing system. A distance sensor is attached to the tip of each arm, and the arm posture is corrected in real time with low computational load, while measuring the distance between the arm tip and the ground during the final moments of the landing operation.

2.2. Upper Side of the Airframe

In some high altitude tasks such as the inspection of the underside of bridges and tunnel roofs, it is advantageous to consider the workspace to be above the robot, making an upper side workspace configuration of the manipulator more suitable.

Examples of aerial manipulators where the workspace is located on the upper side of the airframe can be found in [32–39]. In [32], an aerial manipulation in which the robotic manipulator is attached on the upper side of the airframe with the intention to contact the backside surface of a structure is presented. In [33], an upper side mounted gripper was used for perching to a bar-like object. In [34], a multirotor UAV is equipped with a three degree-of-freedom manipulator on the upper side of the airframe, and exhibited contact force control for inspecting the backside of a bridge.

A contribution towards this kind of aerial manipulator was made by our research group with [40] (**Fig. 3**). A robotic hand mounted on the upper side of a quadrotor allows to grab a rod-shaped object on the upper side of the aircraft while flying. The operation of the aerial platform



Fig. 3. Aerial manipulator with upper side workspace [40] (©2015 IEEE).

itself is a manual, as the operator moves the machine toward the rod to be gripped. In order to grab the rod, it is normally necessary to precisely position the UAV, so that the rod-shaped object fits inside the open hand, hence requiring a positioning accuracy within only a few cm. It is difficult for the robot operator to achieve this by manually controlling the position of the aircraft. Therefore, we installed a camera on the side of the gripper facing upward so that the relative position of the rod-shaped object in relation to the gripper can be observed in the video feed. In addition, a horizontal slide mechanism has been added to the hand part so that the whole hand assembly at the top of the airframe can autonomously move for alignment. The translational movement of the hand is controlled by onboard visual feedback processing. In order to control the hand position with low delay and high speed against the shaking of the aircraft, as well as reducing the strain on the available payload, a FPGA was used for the detection of the rod position by image processing and the motor position control.

Furthermore, apart from grasping the object with a gripper mounted on the top of the airframe, we examined in another project the capability of aerial twisting work by applying torque to an object around its vertical axis [41]. Considering a specific task, we tried to twist a light bulb and unscrew it from its socket. A multirotor aerial robot can generate yaw rotation motion by generating counter torque by balancing the rotation direction of the propeller. The initial problem in this work was how to generate the torque needed to remove the bulb from the socket. While the yaw torque produced of this particular quadrotor is about 0.5 Nm, a torque of about 1 Nm is required to initially loosen a light bulb. The problem was solved by incorporating a lightweight impact mechanism that can abruptly generate a large torque by using the strong force when the aerial platform frame in yaw motion rotates and collides with a custom designed impact component between the aircraft and the hand. With this mechanism, we were able to generate a short burst torque of about 4 Nm, enough to loosen and consequently unscrew various light bulbs.

We have also developed an aerial robot that per-



Fig. 4. Multirotor UAV with upper side mounted omni-wheels for work on the ceiling requiring precise movement [42].

forms work while positioning itself with high accuracy at a specified position on the planar ceiling surface for maintenance and inspection work of various structures (Fig. 4) [42]. In order to improve the positioning accuracy compared to the hovering state, omni-wheels are mounted on the upper part of the airframe, enabling the aerial platform to do high precision locomotion very close to a ceiling surface by using its wheels. The propeller thrust pushes the aircraft against the ceiling surface and are controlled via onboard processing of integrated force sensors to generate just enough friction force for the wheels to propel the aerial platform, in contrast to maxing out the propeller speed of the aerial robot, to conserve energy and enhance the locomotion positional accuracy. Two cameras were mounted downward on the lower part of the aircraft, observing a marker that defines the coordinate system of the aerial platform. The position of the aircraft on the ceiling in relation to the marker is measured to position it at a targeted ceiling position within sub cm accuracy.

2.3. Lateral Side of the Airframe

Having the manipulator positioned on the lateral side of the airframe poses very unique challenges. To allow a VTOL platform to move back and forth and left and right, the propeller usually needs to be tilted in the desired direction to accelerate. Since in case of a multirotor platform each propeller is usually fixed to the frame, it is necessary to tilt the entire aircraft. This change in frame-posture due to underactuation makes it more difficult to control work in the front-back and left-right directions than work in the vertical direction and requires active tilting or passive compliance of the manipulator to be able to counter the frame posture movement during the manipulation task. Another aspect making lateral positioning of the manipulator problematic is the requirement for multirotor platforms to have the center of mass located at the center of the airframe. Putting a manipulator on the lateral side hence requires either a counterweight solution, or an extremely light frame construction of the manipulator. Nonetheless it can be advantageous to have a workspace

next to the multirotor body, on its lateral side. Several researchers have worked around the mentioned challenges and presented solutions for work done on vertical surfaces such as the walls of buildings. Some examples of studies on aerial manipulator with a workspace located on the lateral side of a multirotor airframe include [43–55].

2.4. Further Down Side of the Airframe

Due to payload restrictions, the reach of robotic arms mounted to an aerial platform is generally not very long. This usually results in a workspace set relatively close to the UAV itself. However, some applications aim for a workspace located far away from the UAV body. For instance, let's consider a desired graspable object is very lightweight or placed unstably beneath the aerial robot. In this case, when trying to pick it up or otherwise try to manipulate the object, the effect of downwash, i.e., the downward wind generated by the propellers of the multirotor, can no longer be ignored. In previous research, we have achieved airborne docking of two multirotors using a winch mechanism and a wire-suspended holding bar, countering the effects of downwash of the upper located UAV towards the lower one [56]. Another viable application for far reaching manipulation is when the target is in a narrow region or cluttered environment. In these cases, it might be difficult to approach the target, in order to initiate manipulation, for a multirotor UAV directly. [57] shows the development of a search-and-rescue aerial manipulator. Since the platform is supposed to be used in close cooperation with human rescue workers or to establish physical contact with humans, it is considered essential to maintain a safe distance from such targets to the main aerial platform frame, mounting the propellers.

Several aerial manipulators have been proposed that allow the workspace to be located further away from the airframe by mounting a lightweight long arm on an airframe [58–63]. An interesting outlier is the research in [64–66]. Instead of attaching a manipulator directly to an airframe, a large manipulator is attached to the underside of an active platform with 8 thrusters suspended by a cable attached to a large structure. While technically an aerial manipulator, the lack of a mobile airframe unfortunately results in a severely limited deployment distance. We also considered a similar approach for long-reach manipulation using a wire-suspended device (**Fig. 5**) [67,68]. We found that in order to sufficiently reduce the effect of downwash from an aerial frame transporting the device, a distance of about 5 m for a hexarotor with a width of 80 cm is required [56]. By attaching a robotic gripper from the aircraft with a wire, this distance was achieved while keeping the mechanism as lightweight as possible. Ducted fans, controlled according to onboard analysis of IMU sensor data, were mounted on the gripper to suppress wire swing. This also enabled the dynamic control of fine movement and posture of the robotic gripper. The gripper was also equipped with a downward camera, which was used to detect objects onboard during pickup tasks to autonomously control the attached gripper.



Fig. 5. Wire-suspended long-reach aerial manipulator [67] (©2019 IEEE).

3. Force and Torque for Aerial Work

Motion of the multirotor aerial robot is achieved by changing the rotational speed of its motors. Although an underactuated system, by controlling roll, pitch, yaw and throttle we are able to move in all directions of the three-dimensional space. For the following considerations, a hexarotor aerial platform is used as example. Every two of the rotors that are opposed to each other turn clockwise and counterclockwise, respectively. Theoretically, if there would be no disturbance and they all turn the same speed, the system flies straight up, once the sum of thrust overcomes the weight of the aerial platform's frame. But, since there are always disturbances due to wind, air-frame vibration, or small imbalances, the system state needs to be constantly measured with a high precision inertial measurement unit (IMU) and motor speeds need to be constantly readjusted [69, 70].

The total thrust T is the sum of the thrust f_i from i -th rotor:

$$T = \sum_{i=1}^N f_i = c_T \sum_{i=1}^N \omega_i^2 \quad (1)$$

where c_T is the thrust coefficient that depends on rotor geometry and profile, ω_i is the rotational speed of i -th rotor, and N is the number of rotors.

In hover state, the thrust is equal to the weight of the frame (**Fig. 6** (left)).

$$T - mg = 0 \quad (2)$$

In order to apply force to the external object, larger thrust is required. In **Fig. 6** (right), the multirotor makes contact with the wall and ceiling to apply force. Here, the thrust T is divided into force in vertical T_y and in horizontal T_x ,

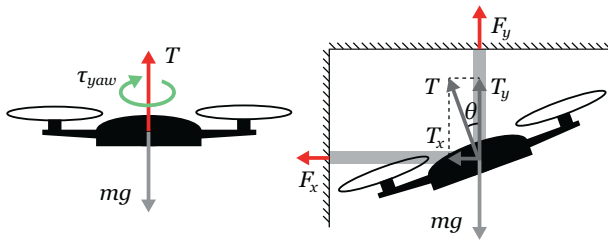


Fig. 6. Force and torque for aerial work. In hover state (left) and in physical contact with environment (right).

and upward force F_y and horizontal force F_x that can be used for manipulation work are

$$F_y = T_y - mg = T \cos \theta - mg \quad (3)$$

$$F_x = T_x = T \sin \theta. \quad (4)$$

In this way, a force obtained by subtracting the own weight from the total thrust can be applied upward, the same force as the own weight can be applied downward, and a force determined according to the tilt angle of the aircraft can be applied in the horizontal direction.

For yaw rotation, an i -th rotor generates a reaction torque τ_i acting on the airframe, and the relationship between reaction torque τ_i and rotational angular velocity ω_i of i -th rotor can be modeled as

$$\tau_i = c_Q \omega_i^2 \quad (5)$$

where c_Q is the coefficient determined by the area of rotational surface, the rotational radius and air density [69]. The sum of reaction torque from six rotors are calculated with

$$\tau_{yaw} = c_Q (\omega_1^2 + \omega_3^2 + \omega_5^2) - c_Q (\omega_2^2 + \omega_4^2 + \omega_6^2). \quad (6)$$

If we assume that $\omega_1 = \omega_3 = \omega_5 = \omega_{CW}$ and $\omega_2 = \omega_4 = \omega_6 = \omega_{CCW}$, the torque in yaw rotation of the airframe can be written as

$$\tau_{yaw} = 3c_Q (\omega_{CW}^2 - \omega_{CCW}^2). \quad (7)$$

We can see that the larger the differences of rotational angular velocity, the larger the reaction torque can become for torsional manipulation.

In several studies, an impulsive force is used in order to increase the maximum torque generated with the multirotor aerial robot. When the velocity changes from v to $v' (= 0)$ in the time duration Δt , averaged impulsive force F_p can be written as

$$F_p = m \frac{v' - v}{\Delta t} \quad (8)$$

where m is mass of the aerial robot. Torque with the impulsive force T_p is

$$T_p = F_p r = m \frac{v' - v}{\Delta t} r \quad (9)$$

where r is the radius of the point of application of force. In order to increase the maximum force, r , m , and v should be increased and Δt should be decreased.

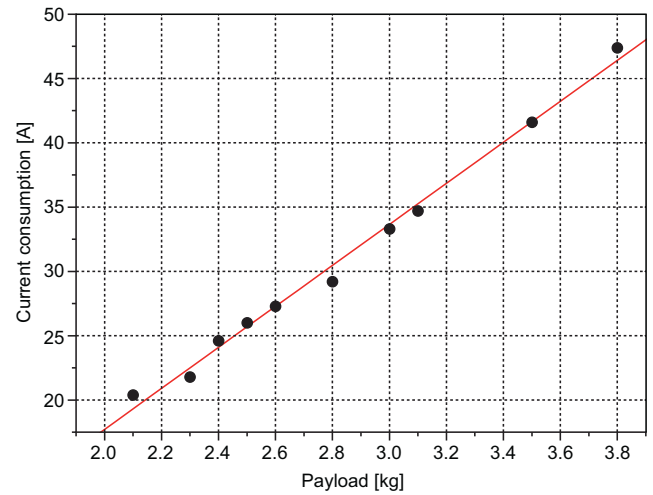


Fig. 7. Experimental result on total current consumption of a hexarotor aerial robot for different payloads in hover state. Weight of the hexarotor is about 2.1 kg, steady voltage of 14.8 V was supplied via cable, and additional load up to 1.7 kg was attached. The linear approximation is $y = 15.9x - 14.2$.

As mentioned in a previous section, in [41] we proposed a torsional manipulation with yaw rotation of a hexarotor aerial robot equipped with a gripper on top of the airframe, with the aim of extending the range of high altitude tasks using aerial manipulation. By introducing an impact mechanism, we were able to increase maximum torque to a value more than six times regular torque. This amount of torsional force allowed the aerial robot to detach a light bulb from its socket during flight. Another example of utilization of aerial impulse impacts is shown in [44]. Here, proper adhesion to the environment is achieved by perching onto smooth vertical surfaces using an adhesion method based on passive vacuum cups with impulse impact capability.

Since usually for multirotor platforms fixed angle propeller blades are used, it is necessary to increase the rotation speed of the rotor, in order to generate a large thrust force. This results in an increase of power consumption of the motor and thereby shortens the deployment time. **Fig. 7** shows the experimental results on the relationship between payload and current consumption for the hexarotor frame used in [42]. When a 1.7 kg payload is added, more than twice the current consumption is required compared to the case of a multirotor aircraft alone. Usually a battery is mounted in the aerial platform to ensure a large range and flexibility in reaching the desired workspace's position. If the desired task workspace is limited and narrow, as well as located close to the ground, power can potentially be supplied by wire as in [42, 71, b].

4. Perching

Several challenges regarding aerial manipulation have been mentioned in previous parts of this work. These



Fig. 8. Aerial manipulator with upper side mounted gripper for perching and underside mounted manipulator with underside, upper and lateral side workspace [79] (©2018 IEEE).

challenges include the low positional holding accuracy of the airframe in relation to the workspace, shaking of the machine body due to propeller stress, and because of the overall small payload capability a limit of the battery capacity and hence short deployment time. If the multirotor body is fixed to a well grounded object without the need for constant hovering, these restrictions can be alleviated. This brings us to the topic of perching for aerial vehicles.

There are several researches on perching on the underside [72–77], upper side [33, 35, 78, 79], or lateral side [47, 80, 81] of the airframe. In [75], mechanisms specific for UAV perching on fence posts in an agricultural setting were explored. In [78], magnetic adhesion was used for perching on the ceiling. The UAV in [81] perches on vertical wall surface using a mechanism with two spiny feet and tail. An aerial manipulator presented in [82] has perching function, knob-twisting function, and door-pushing function to do door opening task. In [83], a micro UAV adheres to the ground and then uses a winch to pull heavy objects. By using this strategy, the UAV is able to tug up to 40 times their mass while adhering to a surface.

In [79], we proposed a robot that holds on to an object near a certain workspace by perching an object with a gripper mounted on the upper part of the airframe and is then able to do manipulation tasks with another manipulator mounted to its underside (**Fig. 8**). Since the propeller can be stopped after the aircraft is fixed, battery consumption is reduced and it is possible to continuously work with the manipulator on the same task for a prolonged time. Furthermore, since the propellers are stopped, the robot overcomes a general limitation of aerial manipulators and can reach into upper and lateral side workspace with a underside attached manipulator. The perching mechanism on the upper side of the airframe has one degree of freedom of rotation that is the yaw direction. Together with the robot arm attached to the underside of the airframe that employs two degrees of freedom, the complete system creates a 3-axis articulated robot when perched.

5. Summary and Future Challenges

Aerial manipulation is a very enticing field in current research and industry and is proliferating itself as the next big application of multirotor platforms after aerial photography, surveillance, and transportation. In this paper, many related studies are reviewed and categorized in regards to composition of the workspace and the forces and torques that can be used for various tasks.

Many issues remain for a practical use. First, a reliable position estimation technique is required in order to accurately position the airframe in relation to the desired workspace. Vision-based position estimation systems are promising in environments where GPS cannot be used but are usually computationally complex, making onboard processing a challenge. Second, more reliable teleoperation techniques are needed to perform complex tasks within aerial manipulation. In addition to the movement of the UAV itself (4 degree of freedom), additional degrees of freedoms of the manipulator are added. This makes a control interface that is easy for an operator to handle in areas where full autonomy is impossible, i.e., nearly every uncontrolled environment, a requirement. It is possible to reduce workload and decision making stress on the human operator by partially incorporating autonomous control, but that again raises the computational complexity to solve a certain task.

One of the advantages of the typical multirotor is the mechanical simplicity of the frame structure, which, generally speaking, consists of motors and propellers fixed to a stiff frame. One disadvantage of this simple structure is the fact that the degrees of freedom that can be independently controlled in the three-dimensional space are limited to horizontal and vertical movement as well as rotation in the yaw direction. That is, the aircraft underactuated and therefore can not be tilted while maintaining its position. In order to eliminate this restriction and increase the degree of freedom, research is being conducted by several research groups on a multirotor that can take any position and posture. One way to achieve this is a tilt rotor mechanism capable of tilting the propellers of a multirotor machine. With this, the posture can be changed significantly while maintaining the position of the aerial robot body [84–87, c]. For example, the Voliro [86], developed at ETH Zurich, allows each propeller to rotate around the frame arm axis and take any position. It also enables the posture of the main body to be vertical or turned over. This makes it possible to perform contact inspections on structures at various angles [c]. Such a multirotor with multiple degrees of freedom can be used for expanding the work range by a manipulator attached to an airframe in various applications of contact tasks.

We expect that the number of aerial manipulators that can generate the necessary forces to perform various aerial manipulation tasks will further increase in the future. This is likely due to the recent development of related technologies such as onboard processors, sensors, control and teleoperation strategies in addition to improvements of the underlying mechanism, such as more

efficient motors or high capacity, lighter batteries. When considering the future outlook of these technologies and their rapid development, there is a lot exciting future development to look forward to in the field of aerial manipulation.

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